3. **Vitamin D:** Vitamin D promotes absorption of calcium.

4. **Sex hormones:** Sex hormones seem to have an effect on calcium and phosphorus balance.

5. **Lactic acid:** Lactic acid produced by microbial fermentation of sugars in the gut,increases the solubility of calcium salts, and increases their absorption.

6. **Fatty acids:** When the absorption of fats is impaired, calcium salts of fatty acids are formed to a greater extent retarding the absorption of calcium.

7. **Oxalates:** Oxalic acid from vegetables like cabbage, and spinach forms insoluble calcium oxalates, thus lowering the calcium absorption.

8. **Phosphates:** Excess of phosphate in diet lowers the calcium absorption. The ratio of calcium and phosphorus in the diet should be 1:1 for optimal absorption of calcium.

9. **Phytic acid:** Phytic acid present in cereals decreases absorption of calcium.

Excretion:

The excretion of calcium is partly through the kidneys but mostly via small intestine. Small amount of calcium may also be lost in sweat.

Deficiency:

Calcium deficiency in body causes the following disorders.

Osteoporosis

Osteoporosis is characterized by demineralization of bones (bone resorption) resulting in progressive loss of bone density.

Tetany

Tetany is a condition characterized by neuromuscular irritability and convulsion, which is associated with a state of hypocalcaemia.

Rickets

Rickets is more directly related to vitamin D deficiency, but calcium and phosphorus metabolism are also involved.

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